VALIDATION WORKFLOW

From a decode hotspot to a measured result

A constrained, reversible optimization with separate correctness and performance gates.

01 PIN
Source & workload
SGLang v0.5.9
Qwen2.5-0.5B BF16

→

02 MEASURE
Native baseline
deterministic Triton
fixed requests

→

03 PROFILE
Inspect path
Nsight Systems
matmul geometry

→

04 ADAPT
Small-M route
opt-in / guarded
SGLang #42145

→

05 VERIFY
Exact outputs
GEMM / reorder
greedy token IDs

→

06 COMPARE
Serving A/B
5 interleaved pairs
paired +72.4%

CONTROLLED COMPARISON

Original and candidate use the same fixed workload
256 requests × 128 output tokens per arm
AB / BA / AB / BA / AB · paired relative gain

Evidence layers remain independently checkable
Exact oracle + recorded kernel launch path
Paired serving results + ablation records